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RISK FACTORS FOR NEOVASCULAR GLAUCOMA AFTER CARBON ION RADIOTHERAPY OF CHOROIDAL MELANOMA USING DOSE–VOLUME HISTOGRAM ANALYSIS

Naoki Hirasawa, M.D.,*† Hiroshi Tsuji, M.D., Ph.D.,* Hitoshi Ishikawa, M.D., Ph.D.,* Hiroko Koyama-Ito, Ph.D.,* Tadashi Kamada, M.D., Ph.D.,* Jun-Etsu Mizoe, M.D., Ph.D.,* Yoshiyuki Ito, M.D., Ph.D.,† Shinji Naganawa, M.D., Ph.D.,† Yoshitaka Ohnishi, M.D., Ph.D.,‡ Hirohiko Tsujii, M.D., Ph.D.*

*Research Center for Charged Particle Therapy, National Institute of Radiological Sciences (NIRS), Chiba, Japan; †Department of Radiology, Nagoya University Graduate School of Medicine, Nagoya, Japan; ‡Departments of Ophthalmology, Wakayama Medical University, Wakayama, Japan

Purpose: To determine the risk factors for neovascular glaucoma (NVG) after carbon ion radiotherapy (C-ion RT) of choroidal melanoma.

Methods and Materials: A total of 55 patients with choroidal melanoma were treated between 2001 and 2005 with C-ion RT based on computed tomography treatment planning. All patients had a tumor of large size or one located close to the optic disk. Univariate and multivariate analyses were performed to identify the risk factors of NVG for the following parameters; gender, age, dose–volumes of the iris-ciliary body and the wall of eyeball, and irradiation of the optic disk (ODI).

Results: Neovascular glaucoma occurred in 23 patients and the 3-year cumulative NVG rate was 42.6 ± 6.8% (standard error), but enucleation from NVG was performed in only three eyes. Multivariate analysis revealed that the significant risk factors for NVG were $V50_{IC}$ (volume irradiated ≥50 GyE to iris-ciliary body) ($p = 0.002$) and ODI ($p = 0.036$). The 3-year NVG rate for patients with $V50_{IC}$ ≥0.127 mL and those with $V50_{IC}$ <0.127 mL were 71.4 ± 8.5% and 11.5 ± 6.3%, respectively. The corresponding rate for the patients with and without ODI were 62.9 ± 10.4% and 28.4 ± 8.0%, respectively.

Conclusion: Dose-volume histogram analysis with computed tomography indicated that $V50_{IC}$ and ODI were independent risk factors for NVG. An irradiation system that can reduce the dose to both the anterior segment and the optic disk might be worth adopting to investigate whether or not incidence of NVG can be decreased with it. © 2007 Elsevier Inc.

Choroidal melanoma, Neovascular glaucoma, Carbon ion radiotherapy, Dose–volume histogram analysis, CT-based treatment planning.

INTRODUCTION

Almost every choroidal melanoma can be successfully treated by proton radiotherapy (PRT) in terms of local tumor control (1–7). However, enucleation usually cannot be prevented in the patients who develop severe ocular morbidity following radiotherapy of large tumor. From this viewpoint, neovascular glaucoma (NVG) is a key toxicity factor that may thwart the patient's hope for preservation of the eye after radiotherapy.

The size and site of the tumor are important factors for the occurrence of NVG in general (7–9), and recently radiation damage to the anterior segment of the eye and the optic disk has been identified as a risk factor for NVG based on dose–volume histogram (DVH) analysis. Furthermore, Rajendran *et al.* (9) suggested the possibility that the risk of NVG may be reduced by adopting multiportal irradiation, which might reduce the dose to these important structures in some patients. This information is undoubtedly most significant in particle radiotherapy for choroidal melanoma. It should be noted, however, that the volumes and locations of the analyzed structures in these studies involved some approximation because the DVH analyses were based on treatment planning using the eye model. To obtain more reliable information for optimizing treatment planning,

Reprint requests to: Naoki Hirasawa, M.D., Research Center for Charged Particle Therapy, National Institute of Radiological Sciences (NIRS), 4-9-1 Anagawa, Inage-ku, Chiba-city, 263-8555 Japan. Tel: (+81) 43-251-2111; Fax: (+81) 43-256-6506; E-mail: naoki_h@nirs.go.jp

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